

Simulation-Based Inference

Bayesian Simulation Studies

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Simulation Studies: Overview

- Choose a data model (likelihood) $p(y \mid \theta)$
- Choose a prior $p(\theta)$ for θ .
- Choose the number of simulation trials S .
- Simulate S parameters $\theta^{(s)} \sim p(\theta)$ from the prior.
- Simulate S datasets $y^{(s)} \sim p(y \mid \theta^{(s)})$ from the likelihood
- Compute samples $\theta^{(s,t)} \sim p(\theta \mid y^{(s)})$ from the posterior (of the same or a different model than the ground truth)
- Compute further relevant statistics such as posterior means, quantiles etc.
- Summarize the metrics across the the S trials appropriately.

Estimation accuracy

As in the frequentist case, but we have to deal with posterior draws.

RMSE Definition 1:

$$\text{RMSE}_1 = \sqrt{\frac{1}{S} \sum_{s=1}^S (\bar{\theta}^{(s)} - \theta^{(s)})^2}$$

where $\bar{\theta}^{(s)} = \sum_{t=1}^T \theta^{(s,t)} / T$ is the posterior mean and $\theta^{(s)}$ is the (true) reference value.

RSME Definition 2:

$$\text{RMSE}_2 = \frac{1}{S} \sum_{s=1}^S \sqrt{\frac{1}{T} \sum_{t=1}^T (\theta^{(s,t)} - \theta^{(s)})^2}$$

Question: Which of these two definitions would you prefer and why?

A Bayesian posterior is calibrated if the posterior intervals have appropriate coverage. For example, 80% intervals are expected to contain the true parameter 80% of the time.

If data is generated according to a model M , Bayesian posterior inference with respect to M is calibrated by construction. More formally:

$$q = \int \int \mathbb{I}(\theta^* \in U_q(\theta \mid y)) p(y \mid \theta^*) p(\theta^*) dy d\theta^*$$

where $U_q(\theta \mid y)$ is a posterior region/interval with nominal coverage probability q .

The following slides are inspired by the Stan user manual.

Consider a model with prior $p(\theta)$ and likelihood $p(y \mid \theta)$

First, simulate parameters from the prior:

$$\theta^{(s)} \sim p(\theta)$$

Then simulates data given the parameters:

$$y^{(s)} \sim p(y \mid \theta^{(s)})$$

Parameter and data together are thus sampled from the joint distribution:

$$(y^{(s)}, \theta^{(s)}) \sim p(y, \theta)$$

From Bayes's rule, it follows that for any observed (fixed) data $\tilde{y}$:

$$p(\theta \mid \tilde{y}) \propto p(\tilde{y}, \theta).$$

Therefore, the simulated parameters constitute a draw from the posterior for the simulated data:

$$\theta^{(s)} \sim p(\theta \mid y^{(s)}).$$

Now consider an algorithm that produces a sequence of draws from the posterior given this simulated data,

$$\theta^{(s,1)}, \dots, \theta^{(s,T)} \sim p(\theta \mid y^{(s)}).$$

Because $\theta^{(s)}$ is also distributed as a draw from the posterior, the rank statistics of $\theta^{(s)}$ with respect to $\theta^{(s,1)}, \dots, \theta^{(s,T)}$ should be uniform.

Simulation-Based Calibration (SBC)

Suppose the Bayesian model to test has joint density

$$p(y, \theta) = p(y \mid \theta)p(\theta)$$

SBC works by generating s simulated parameter and data pairs according to the joint density,

$$(y^{(1)}, \theta^{(1)}), \dots, (y^{(S)}, \theta^{(S)}) \sim p(y, \theta).$$

For each simulated data set $y^{(s)}$, use our estimation algorithm to be validated (e.g., MCMC) to generate T posterior draws.

If everything is working properly, will be distributed marginally as

$$\theta^{(s,t)}, \dots, \theta^{(s,T)} \sim p(\theta \mid y^{(s)}).$$

For a simulation s and parameter k , the rank of the simulated parameter among the posterior draws is

$$\begin{aligned} r_k^{(s)} &= \text{rank}(\theta_k^{(s)}, (\theta_k^{(s,1)}, \dots, \theta_k^{(s,T)})) \\ &= \sum_{t=1}^M \mathbb{I}[\theta_k^{(s,t)} < \theta_k^{(s)}]. \end{aligned}$$

That is, the rank is the number of posterior draws $\theta_k^{(s,t)}$ that are less than the simulated draw $\theta_k^{(s)}$.

If the algorithm generates posterior draws according to the posterior, the ranks should be (discretely) uniformly distributed from 0 to T :

$$r_k^{(s)} \sim \text{categorical} \left(\pi_0 = \frac{1}{T+1}, \dots, \pi_T = \frac{1}{T+1} \right).$$

The trinity of practical SBC

SBC works if the following three components match:

- The simulator
- The probabilistic program (model to be trained)
- The posterior inference algorithm (e.g., MCMC)

Failure of SBC indicates a problem in at least one component.

Example model

For illustration, a very simple model will suffice. Suppose there are two parameters (μ, σ) with independent priors:

$$\mu \sim \text{normal}(0, 1) \text{ and } \sigma \sim \text{lognormal}(0, 1)$$

The data $y = y_1, \dots, y_N$ is drawn conditionally independently given the parameters

$$y_n \sim \text{normal}(\mu, \sigma).$$

The joint prior density is thus

$$p(\mu, \sigma) = \text{normal}(\mu \mid 0, 1) \cdot \text{lognormal}(\sigma \mid 0, 1)$$

and the sampling density is

$$p(y \mid \mu, \sigma) = \prod_{n=1}^N \text{normal}(y_n \mid \mu, \sigma)$$

Example continued: Sampling from the posterior

For example, suppose the following two parameter values are drawn from the prior in the first simulation:

$$(\mu^{(1)}, \sigma^{(1)}) = (1.01, 0.23).$$

Then data $y^{(1)} \sim p(y \mid \mu^{(1)}, \sigma^{(1)})$ is drawn according to the sampling distribution. Next, $T = 4$ draws are taken from the posterior $\mu^{(1,t)}, \sigma^{(1,t)} \sim p(\mu, \sigma \mid y^{(1)})$:

t	$\mu^{(1,t)}$	$\sigma^{(1,t)}$
1	1.07	0.33
2	-0.32	0.14
3	-0.99	0.26
4	1.51	0.31

Example continued: Computing SBC ranks

Then the comparisons on which ranks are based look as follows:

t	$I(\mu^{(1,t)} < \mu^{(1)})$	$I(\sigma^{(1,t)} < \sigma^{(1)})$
1	0	0
2	1	1
3	1	0
4	0	0

The ranks are the column sums, $r_{\mu}^{(1)} = 2$ and $r_{\sigma}^{(1)} = 1$

Because the simulated parameters are distributed according to the posterior, these ranks should be distributed uniformly between 0 and T , the total number of posterior draws.

Example: Simulator in R

It is usually advisable to write the simulator separately from the probabilistic program.

```
simulator <- function(N) {  
  mu <- rnorm(1, mean = 0, sd = 1)  
  sigma <- rlnorm(1, meanlog = 0, sdlog = 1)  
  y <- rnorm(N, mu, sigma)  
  return(list(N=N, mu=mu, sigma=sigma, y=y))  
}
```

Example: Probabilistic Program in Stan

```
data {  
  int<lower=1> N;  
  vector[N] y;  
}  
parameters {  
  real mu;  
  real<lower=0> sigma;  
}  
model {  
  mu ~ normal(0, 1);  
  sigma ~ lognormal(0, 1);  
  y ~ normal(mu, sigma);  
}
```

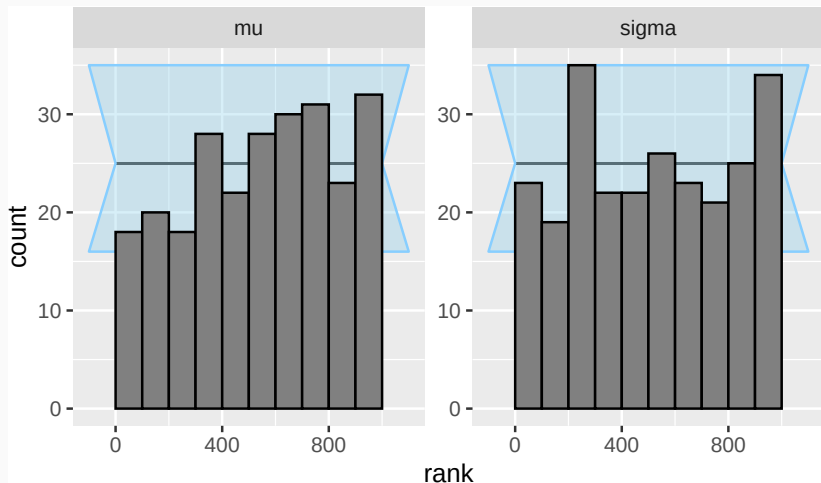
Running SBC

```
library(cmdstanr)
library(posterior)

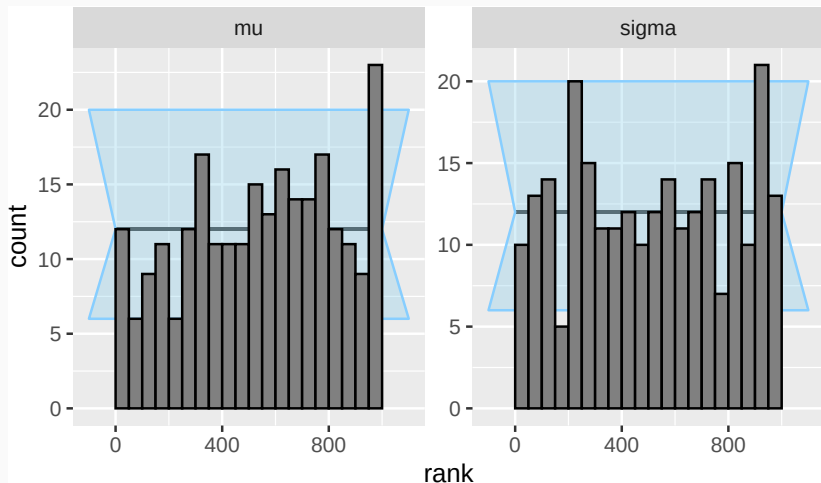
mod_normal <- cmdstan_model(stan_file = "normal_model.stan")

set.seed(62435)
S <- 250
N <- 10
ranks <- data.frame(mu = rep(NA, S), sigma = rep(NA, S))
for (s in 1:S) {
  sdata <- simulator(N = N)
  fit_normal <- mod_normal$sample(data = sdata, chains = 1)
  draws <- fit_normal$draws() |> as_draws_df()
  ranks$mu[s] <- sum(draws$mu < sdata$mu)
  ranks$sigma[s] <- sum(draws$sigma < sdata$sigma)
}
```

Rank Histograms



Rank Histograms: More Bins



The result of SBC checked via histograms may depend on the bins.

Empirical cumulative distribution function (ECDF)

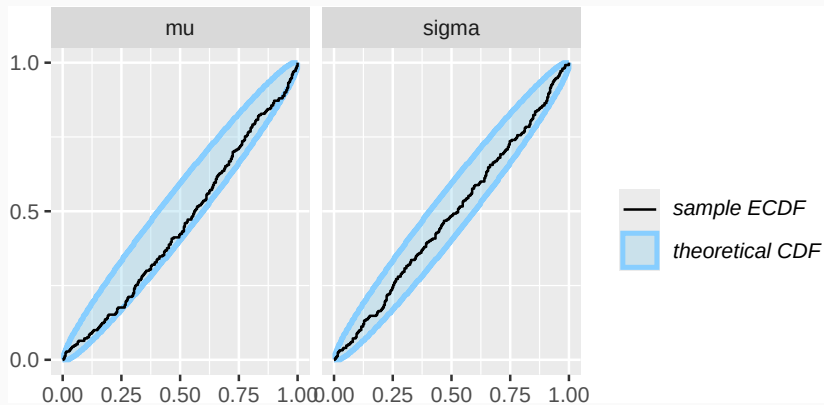
Reminder: The cumulative distribution function (CDF):

$$\text{CDF}(\tilde{x}) = P(x \leq \tilde{x}) = \int_{-\infty}^{\tilde{x}} p(x) dx$$

Estimate the CDF from a finite set of samples $\{x^{(s)}\}_{s=1}^S$ via the ECDF:

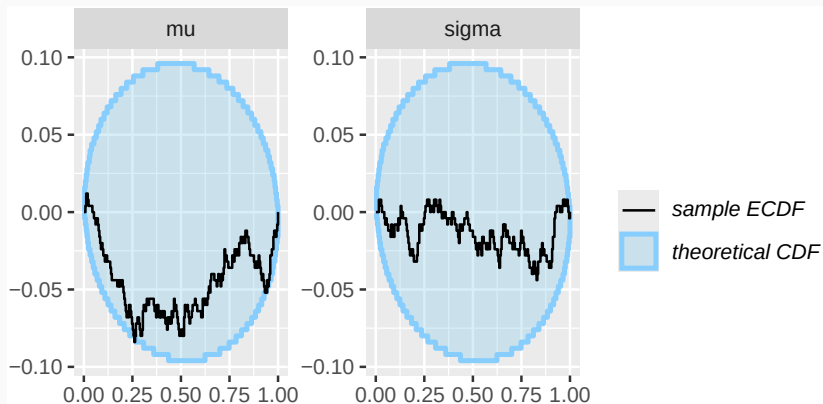
$$\text{ECDF}(\tilde{x}) = P(x^{(s)} \leq \tilde{x}) = \frac{1}{S} \sum_{s=1}^S \mathbf{I}(x^{(s)} \leq \tilde{x})$$

Rank ECDFs

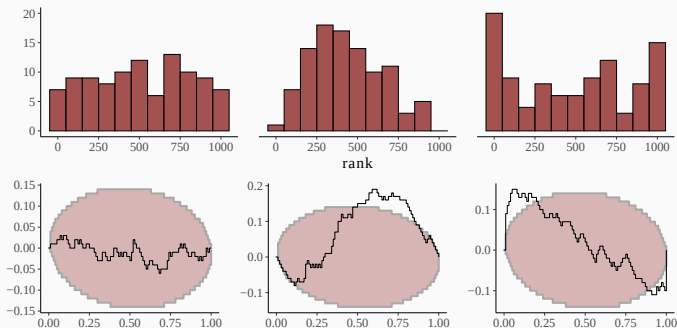


Rank ECDF Differences

Equivalently plot the ECDF-difference $\text{ECDF}(\tilde{x}) - \tilde{x}$



Different kinds of miscalibration



- Left: A well-calibrated quantity.
- Center: A miscalibrated quantity with too many lower ranks indicating a positive bias in the approximate posterior.
- Right: A miscalibrated quantity with too many extreme ranks indicating an overconfident approximate posterior.

Distributions that pass SBC

Not only the true posterior $p(\theta \mid y)$ passes SBC but also

- The prior $p(\theta)$
- Any linear combination of prior and posterior $\alpha p(\theta) + (1 - \alpha)p(\theta \mid y)$
- For finite number of tested quantities: Many other distributions (which are hard to construct though)

Solution: Test more than just the parameters θ with SBC but also data-dependent quantities $f = f(\theta, y)$ with ranks

$$r_f^{(s)} = \text{rank} \left(f(\theta^{(s)}, y^{(s)}), (f(\theta^{(s,1)}, y^{(s)}), \dots, f(\theta^{(s,T)}, y^{(s)})) \right)$$

Example: The joint log likelihood $f(\theta, y) = \log p(y \mid \theta)$

SBC vs. data-averaged posterior checks

Bayesian inference has multiple self-consistency properties.

Data-averaged posterior (DAP) = prior:

$$p(\theta) = \int \int p(\theta | y) p(y | \theta^*) p(\theta^*) dy d\theta^*$$

So if we would run the SBC procedure and then put all posterior draws of all model runs together, the resulting posterior histogram should look like the prior.

Important: DAP is not SBC!

- DAP checks marginal calibration
- SBC checks conditional calibration
- SBC is usually the more powerful check

Summary

Bayesian simulation studies proceed in much the same way as their frequentist counterparts. Except that we do not fix the true parameter values but rather fix the prior from which parameters are then sampled.

When summarizing the results of Bayesian simulation studies, care must be taken how to average both *across* the simulation trials and posterior draws *within* each trial.

SBC is a powerful method to test for the correctness of probabilistic programs and posterior approximators (e.g., MCMC or VI).

Limitations of SBC include:

- It is computationally expensive.
- Also some distributions other than the true posterior can pass SBC.
- Since simulator and probabilistic program must match, we cannot check model-misspecified scenarios with SBC.